11 Chemistry
Unit 2 Topic Test 1: Concentration, Standard Solutions and Dilution

Details	
Student Name:	Teacher Name:
Date Set: ____/____/26	Due Date: ____/____/26

Curriculum Assessed

On completion of Unit 2 the student should be able to measure and analyse chemicals in water. This topic test assesses concentration in mol L^{-1}, g L^{-1} and % (m/v) and conversions between these units, the preparation of standard solutions, and dilution.

Task Overview

Section A – Multiple Choice: 10 questions, 1 mark each (10 marks). Answer on the separate multiple-choice answer sheet.
Section B – Short Answer: 4 questions (30 marks). Answer in this booklet.

Conditions

- Test conditions (revision topic test)
- VCE Chemistry Data Book provided
- Permitted materials: pens, pencils, highlighters, erasers, sharpeners, rulers and one scientific calculator (no CAS).
- Not permitted: notes of any kind, blank paper, white-out, mobile phones, smart watches or any other electronic device.
- 46 mins writing time (about 1.15 minutes per mark)

Marking

- 40 marks
- Marked according to teacher-made marking guide.
- Show all working in calculation questions; give final answers with units and an appropriate number of significant figures. Equations must be balanced and include states.

A+	A	B+	B	C+	C	D+	D	E+	E	NS
90-100%	80-89%	75-79%	70-74%	65-69%	60-64%	55-59%	50-54%	45-49%	40-44%	<40%

Total Marks:		Percentage:		Letter Grade:	

SECTION A – Multiple-choice questions (10 marks)

Answer **all** questions on the separate multiple-choice answer sheet by filling in the letter of the best response. Each question is worth 1 mark. No marks are deducted for incorrect answers. Do **not** write your Section A answers in this booklet.

Question 1

A cleaning solution is labelled '2.5% (m/v) ammonium chloride, NH_4Cl'.

The label indicates that there is

A. 2.5 g of NH_4Cl dissolved in every 100 mL of water, H_2O.

B. 2.5 g of NH_4Cl in every 100 mL of solution.

C. 2.5 g of NH_4Cl in every 100 g of solution.

D. 2.5 g of NH_4Cl in every 1.0 L of solution.

Question 2

A student dissolves 3.40 g of sodium nitrate, $NaNO_3$, in deionised water, H_2O, and makes the solution up to 200.0 mL.

The concentration of the sodium nitrate solution is

A. 2.00×10^{-4} M

B. 17.0 M

C. 8.00×10^{-3} M

D. 0.200 M

Question 3

Which one of the following best describes a standard solution?

A standard solution is a solution

A. whose concentration is accurately known.

B. that has a concentration of exactly 1 M.

C. that has been made by diluting a more concentrated solution.

D. that is made at standard laboratory conditions (SLC).

Use the following information to answer Questions 4 and 5.

An eyewash solution contains boric acid, H_3BO_3, at a concentration of 2.0% (m/v).

Question 4

The concentration of boric acid in this solution, in mol L^{-1}, is closest to

A. 0.032 M

B. 3.2 M

C. 0.32 M

D. 1.2×10^3 M

Question 5

A nurse adds 150 mL of water, H_2O, to 50 mL of the 2.0% (m/v) boric acid solution.

Which row correctly describes the diluted solution, compared with the original 50 mL?

	Amount of H_3BO_3 (mol)	Concentration of H_3BO_3 (% (m/v))
A.	unchanged	0.67
B.	unchanged	0.50
C.	decreased	0.50
D.	unchanged	8.0

Question 6

To analyse a nickel electroplating solution, an analyst pipetted 4.00 mL of it into a 200.0 mL volumetric flask and filled the flask to exactly 200.0 mL with deionised water, H_2O. The diluted solution had a nickel(II) ion, Ni^{2+}, concentration of 0.0170 M.

The concentration of Ni^{2+} in the original electroplating solution was

A. 0.850 M

B. 3.40×10^{-4} M

C. 0.833 M

D. 0.867 M

Question 7

Gardeners add aluminium sulfate, $Al_2(SO_4)_3$, to soil to turn hydrangea flowers blue. A gardener's solution has an aluminium sulfate concentration of 0.0200 M.

The concentration of sulfate ions, SO_4^{2-}, in this solution is

A. 0.0200 M

B. 0.0400 M

C. 0.100 M

D. 0.0600 M

Question 8

An energy drink contains caffeine, $C_8H_{10}N_4O_2$, at a concentration of 320 mg L^{-1}.

The amount of caffeine in one 500 mL can of the drink is closest to

A. 1.65×10^{-3} mol

B. 0.825 mol

C. 8.25×10^{-4} mol

D. 3.30×10^{-3} mol

Question 9

A student prepares a standard solution in a volumetric flask. Consider the following statements about the procedure.

I The solid is dissolved in a small volume of deionised water, H_2O, in a beaker before it is transferred into the volumetric flask.

II After the solution is made up to the mark, the flask is stoppered and inverted several times to mix it.

III The volumetric flask may be warmed over a Bunsen burner to make the solid dissolve faster.

Which of the statements are correct?

A. I and II only

B. I only

C. II and III only

D. I, II and III

Question 10

Ammonium sulfate, $(NH_4)_2SO_4$, is used as a lawn fertiliser.

The mass of ammonium sulfate that must be dissolved to make 200.0 mL of a solution in which the concentration of ammonium ions, NH_4^+, is 0.180 M is

A. 4.76 g

B. 23.8 g

C. 2.05 g

D. 2.38 g

END OF SECTION A

SECTION B – Short-answer questions (30 marks)

Answer **all** questions in the spaces provided. Show all working in calculation questions. Give final answers with units and appropriate significant figures. Equations must be balanced and include states.

Question 1 (7 marks)

Jasmine descales her espresso machine with citric acid, $C_6H_8O_7$. The instructions say to dissolve one 25.0 g sachet of citric acid in water, H_2O, and make the solution up to 750 mL.

a) Calculate the concentration of citric acid in this solution, in g L^{-1}.

1 mark

b) Convert your answer to part a) to a concentration in % (m/v).

1 mark

c) Calculate the concentration of citric acid in this solution, in mol L^{-1}.

3 marks

d) For weekly cleaning, the manufacturer recommends 500 mL of a 0.0800 M citric acid solution. Calculate the volume of Jasmine's solution from part c) that must be diluted with water to make this weekly cleaning solution.

2 marks

Question 2 (7 marks)

In the school laboratory, Ngoc, a Year 11 student, needs a standard solution of sulfamic acid, NH_2SO_3H (M = 97.1 g mol^{-1}), to standardise a sodium hydroxide solution. Sulfamic acid is a primary standard.

She weighed the sulfamic acid by difference, dissolved it in deionised water, H_2O, and made the solution up to exactly 500.0 mL. Her weighing results are shown below.

Measurement	Mass (g)
weighing bottle + sulfamic acid, before transfer	18.734
weighing bottle + remaining sulfamic acid, after transfer	13.822

a) Calculate the mass of sulfamic acid that Ngoc transferred.

1 mark

b) Calculate the concentration of the sulfamic acid solution, in mol L^{-1}.

2 marks

c) Explain why sodium hydroxide, NaOH, is not suitable as a primary standard.

2 marks

d) Ngoc's balance had not been zeroed, so every reading was 0.020 g higher than the true mass (the same offset for every reading). Explain whether this error affects the concentration she calculated in part b).

2 marks

Question 3 (7 marks)

Mateo works at a craft brewery in Geelong. He cleans the fermenting tanks with a wash solution of sodium hydroxide, NaOH, that must have a concentration of 2.00% (m/v).

He makes the wash by diluting a stock solution of NaOH that has a concentration of 30.0% (m/v).

a) Convert 2.00% (m/v) to a concentration in g L^{-1}.

1 mark

b) Using your answer to part a), calculate the concentration of the wash solution in mol L^{-1}.

1 mark

c) Calculate the volume of stock solution needed to make 600 L of wash solution.

2 marks

d) After several cleans, the concentration of NaOH in the 450 L of wash in the tank has fallen to 1.40% (m/v). Calculate the volume of the 30.0% (m/v) stock solution that Mateo must add to the tank to bring the wash back to 2.00% (m/v). Assume that the volumes add together.

3 marks

Question 4 (9 marks)

An orchardist in the Goulburn Valley sprays her apple trees with zinc sulfate, $ZnSO_4$, to correct a zinc deficiency. She compares three zinc sulfate concentrates, each labelled in different units.

Concentrate	Label
A	0.0480 M $ZnSO_4$
B	7.50 g L^{-1} $ZnSO_4$
C	0.600% (m/v) $ZnSO_4$

a) Convert the concentration of concentrate C to g L^{-1}.

1 mark

b) Determine which concentrate has the highest concentration of zinc sulfate in mol L^{-1}. Show your working.

3 marks

c) Calculate the concentration of sulfate ions, SO_4^{2-}, in concentrate B, in g L^{-1}.

2 marks

d) The orchardist needs 100 L of spray with a zinc ion, Zn^{2+}, concentration of 0.500 g L^{-1}.
Calculate the volume of concentrate A that she must dilute with water, H_2O, to make this spray.

3 marks

END OF SECTION B

Additional space

Clearly number any response written here.

END OF TEST

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SECTION A – MULTIPLE-CHOICE ANSWER SHEET

Student name: _____ Teacher: _____

Shade or circle ONE letter for each question. Each question is worth 1 mark. If you change an answer, cross out the first answer clearly.

Q	A	B	C	D
1	(A)	(B)	(C)	(D)
2	(A)	(B)	(C)	(D)
3	(A)	(B)	(C)	(D)
4	(A)	(B)	(C)	(D)
5	(A)	(B)	(C)	(D)
6	(A)	(B)	(C)	(D)
7	(A)	(B)	(C)	(D)
8	(A)	(B)	(C)	(D)
9	(A)	(B)	(C)	(D)
10	(A)	(B)	(C)	(D)

Section A score: _____ / 10